Diagnosing Alzheimer's Disease using Early-Late Multimodal Data Fusion with Jacobian Maps

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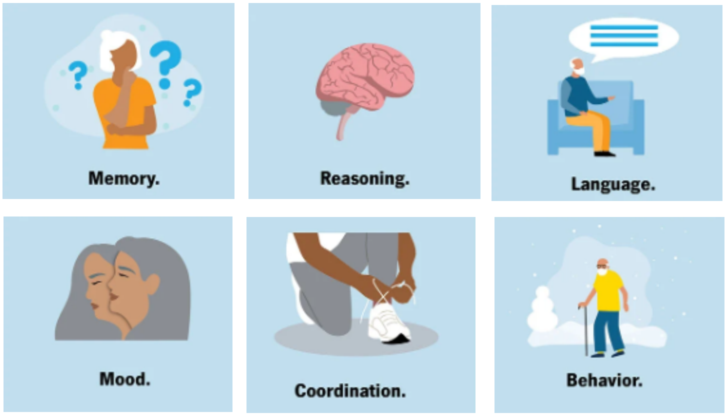
Alzheimer's Disease (AD)

- The lifetime risk for Alzheimer’s disease at age 45:
  1 in 5 women and 1 in 10 men.

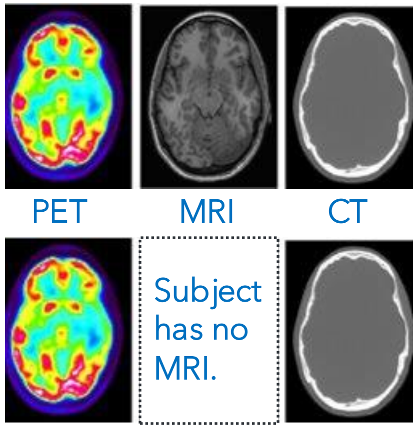
- Kills more people than prostate cancer and breast cancer combined
- The number of people living with AD **doubles** every 5 years, estimated to reach **14 million** by **2060**.
- The symptoms of AD involve a gradual decline in some, most or all of the following:



- Manual analysis of medical scans is challenging
- Challenging to detect early patterns of AD

- Leading to focus on machine learning for early detection and intervention

- **Multimodal Data:**
  - Machine learning with multimodal data has demonstrated promising performance compared to relying solely on a single modality.
- **Missing Modalities:**
  - Incomplete data can **lead to biased** or less accurate model predictions
  - Ignoring subjects with missing modalities can result in a significant loss of valuable information
- **Small Datasets:**
  - Deep models typically perform better
  - But **multimodal AD datasets** are small compared to MNIST
  - leading to **overfitting** where models memorize the training set rather than learning general patterns.
- **Appropriate Preprocessing and Model Selection:**
  - Need to **highlight deformations**
  - Have a model and a **fusion method** that handles the above without sacrificing performance



~1K Subjects
(with missing modalities)

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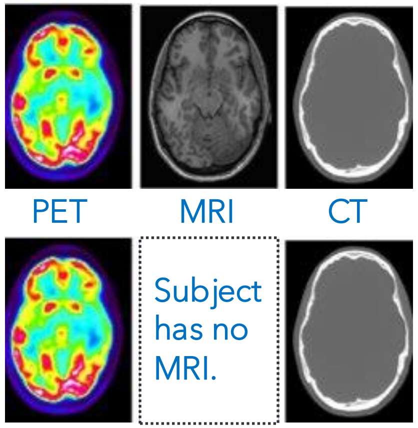

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Jacobian maps

Hot Deck Imputation (HDI)

Early-Late Fusion Model

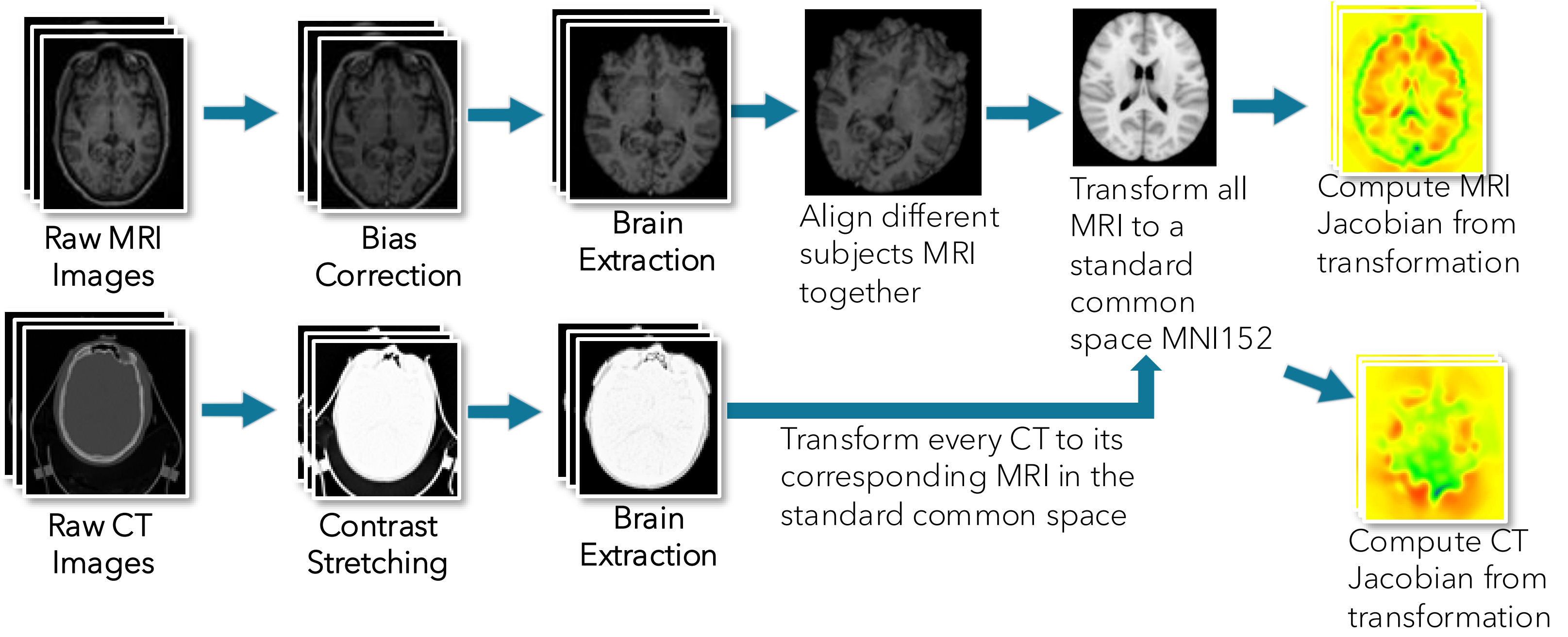
OASIS
OPEN ACCESS SERIES
OF IMAGING STUDIES

- Participants include 755 cognitively normal adults and 622 individuals at various stages of cognitive decline ranging in age from 42-95yrs.
- MRI and CT 3D images
- Based on clinical dementia rating (CDR) scores:

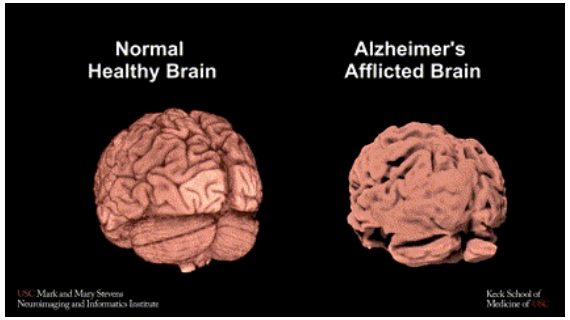
CDR	Class
0	Normal
0.5	MCI
1	Mild
2	Moderate
3	Severe

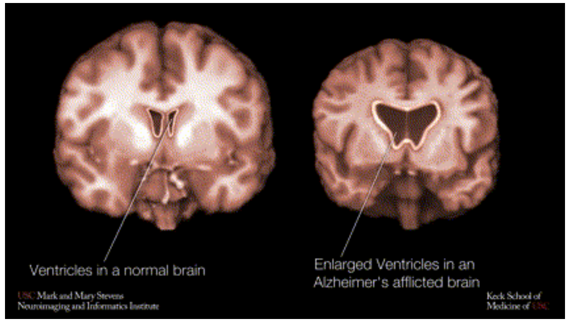
Combined

Preprocessing



Brain Volume Changes Denoting Dementia





Jacobian Maps

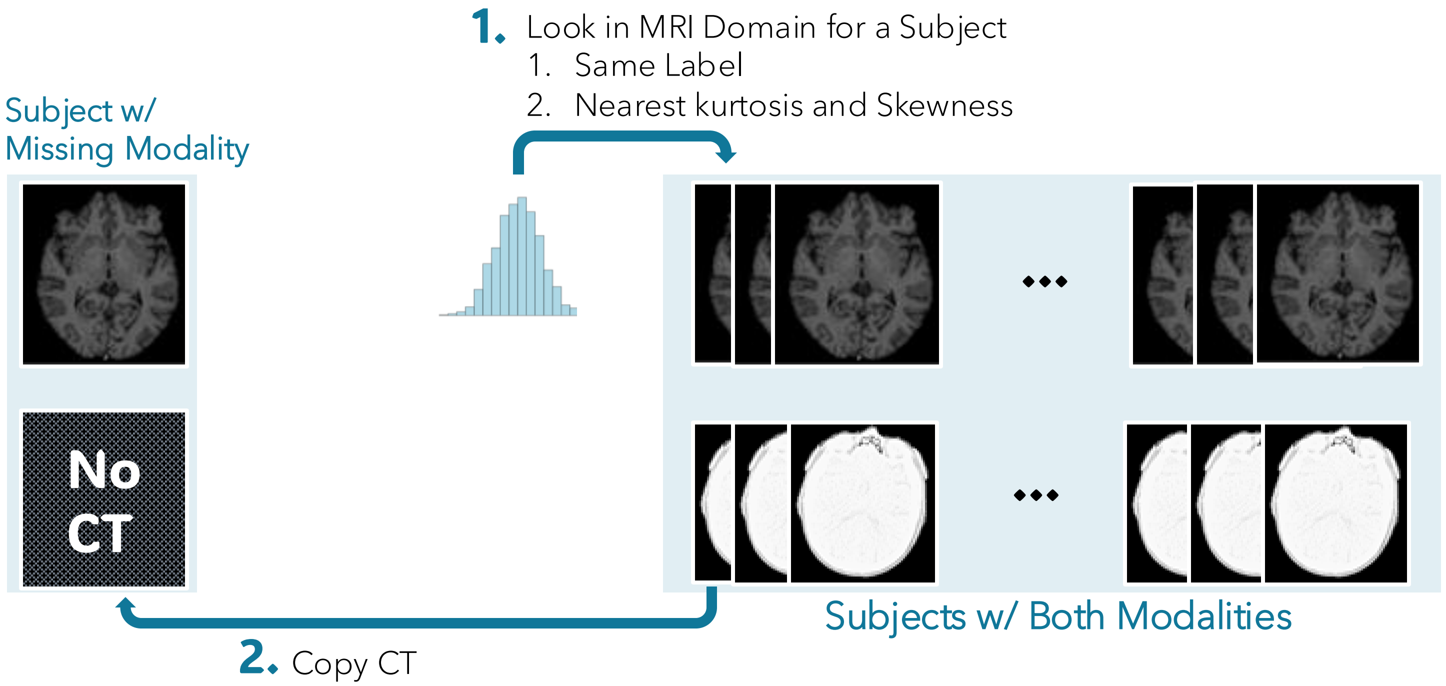
- Capture subtle brain volume changes
- Providing informative representations for feature learning
- Highlight local brain morphometry

$$J(v) = \begin{bmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_x}{\partial y} & \frac{\partial v_x}{\partial z} \\ \frac{\partial v_y}{\partial x} & \frac{\partial v_y}{\partial y} & \frac{\partial v_y}{\partial z} \\ \frac{\partial v_z}{\partial x} & \frac{\partial v_z}{\partial y} & \frac{\partial v_z}{\partial z} \end{bmatrix}$$

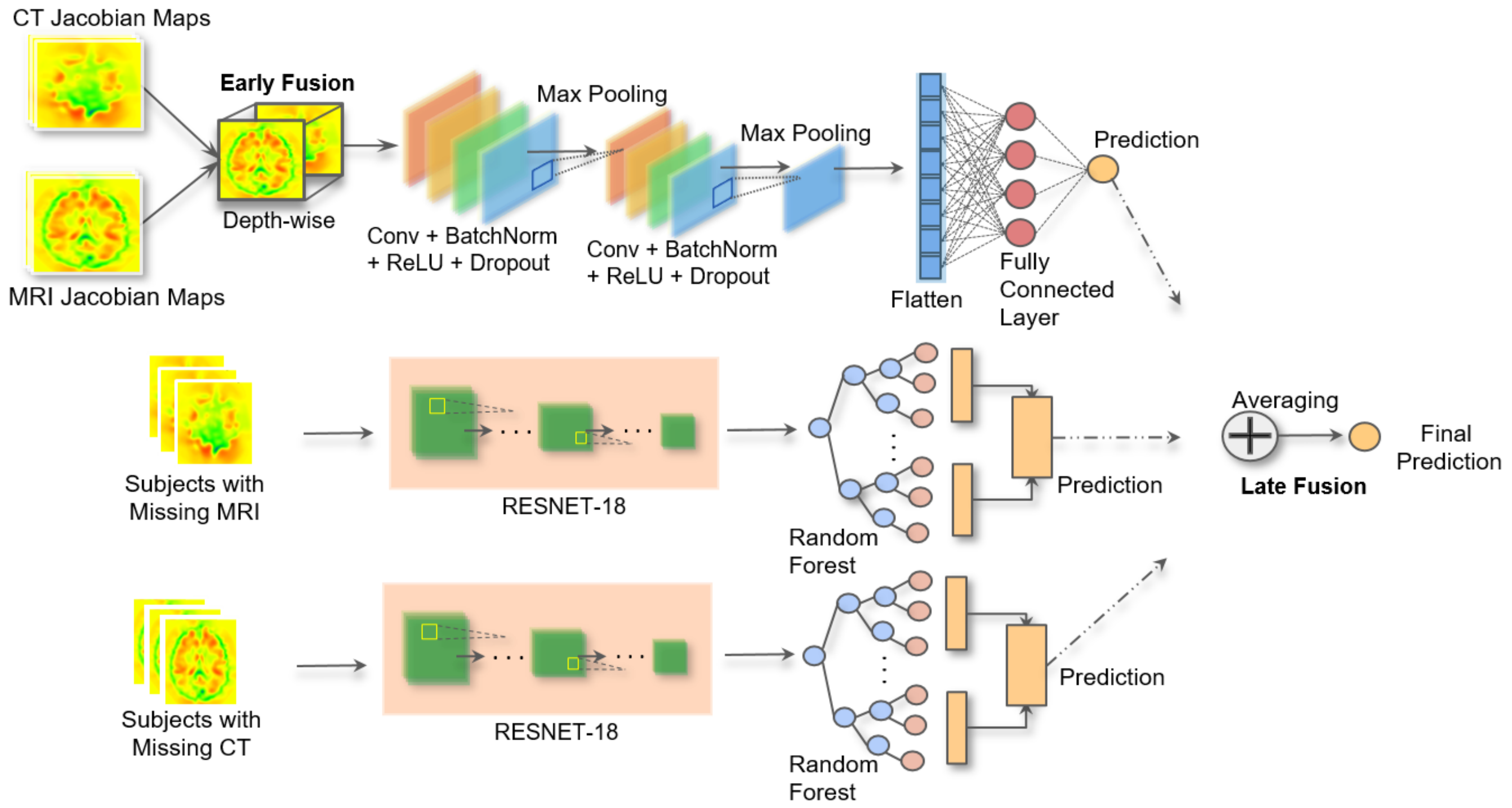
$$J_{map}(M) = \begin{bmatrix} & \vdots & \\ \dots & Det(J(v(x,y,z)) & \dots \\ & \vdots & \end{bmatrix}_{\substack{x=1\dots W \\ y=1\dots H \\ z=1\dots D}}$$

$$\text{At each voxel: } \begin{cases} \text{volume expansion} & \text{if } Det(J) > 1 \\ \text{no change} & \text{if } Det(J) = 1 \\ \text{volume compression} & \text{if } Det(J) < 1 \end{cases}$$

Imputing Missing Values: Hot Deck Imputation

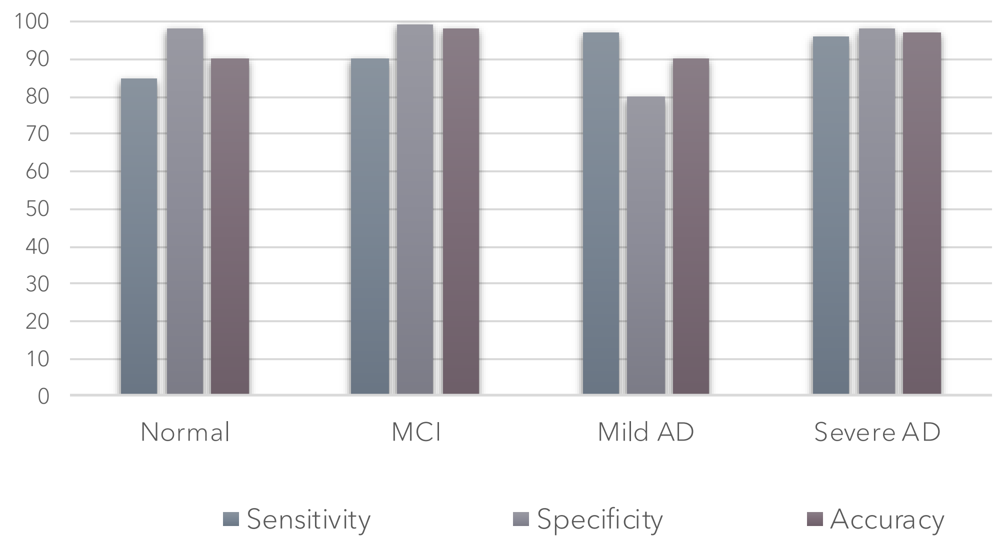


Model Architecture



Performance across Four AD Stages

Model	Accuracy	Sensitivity	Specificity
CNN	91.02	83.37	87.21
RF CT	94.26	86.79	90.52
RF MRI	89.35	83.14	94.47
ELF	**97.19**	**95.19**	**98.76**



Model	Modality	Classes	SENS	SPEC	ACC
Salami et al. 2022	MRI	AD, CN	86	85	87.7
Massalimova et al. 2021	MRI	NC, MCI, AD	96	96	96
Lazli et al. 2019	MRI, PET	AD, healthy	92	91.7	91
ELF	MRI, CT	Normal, MCI, mild AD, severe AD	95.19	**98.76**	**97.19**

Summary

- Introduced Early-Late Fusion (ELF) approach for Alzheimer's disease diagnosis across four stages: normal, MCI, mild AD, severe AD.
- Per-subject registration for alignment among different modalities, Jacobian domain transformation for feature extraction on brain morphometry and shape changes.
- Addressed heterogeneous data modalities using CNN and RF models in ELF framework.
- HDI technique to optimize use of available information
- Achieved a remarkable accuracy of 97.19% in extensive experiments, surpassing recent literature results.
- Contribution towards more effective interventions and treatments for Alzheimer's disease diagnosis in the future.

Thank You